

Vineeth Gutta

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Research Interests

Machine Learning and High Performance Computing (HPC) with a focus on portability, scalability, and performance of Deep Learning (DL) applications running on HPC systems

Technical Skills

Languages: Python, Java, C++, React JS, Node JS, Bash, SQL

Software and Tools: Git, Docker, PyTorch, TensorFlow, Scikit-Learn, Numpy, Pandas, ONNX, ONNX Runtime, REST, Eclipse, relational databases, NoSQL

High Performance Computing (HPC) software: Slurm, Singularity, Horovod, PyTorch Distributed Data Parallel, RAPIDS AI, DeepSpeed, AMD ROCm, GPU Profiling (Omnitrace, PyTorch Profiler, NSight)

Education

Doctor of Philosophy (PhD) in Computer Science

University of Delaware – Newark, DE

Advisor: Dr. Sunita Chandrasekaran

(2019 – Present)

Cumulative GPA: 3.95

Expected: January 2025

Bachelor of Science in Computer Science

University of Delaware – Newark, DE

(2015 – 2019)

Cumulative GPA: 3.43

Research Experience

Computational Scientist: Dr. Sunita Chandrasekaran's Computational Research and Programming Lab

- Research focus is on improving compatibility between evolving DL frameworks and advancing hardware architectures to make it more seamless for ML practitioners to port software and new models to new hardware architectures.

Collaboration with Frederick Nat. Lab for Cancer Research (FNLCR)/ National Cancer Institute (NCI/NIH):

(June 2021 – Present)

- Improved portability of applications like AMPL and CANDLER which leverage DL to automate key drug discovery steps and address cancer surveillance respectively
- Developed a library for assessing the predictive efficacy of cancer drug response models across Convolutional Neural Networks (CNNs) and Gradient Boosted Tree models, particularly beneficial for tabular dataset problems.
- Worked on scaling DL model training on AMD MI250X GPUs and implemented hyperparameter optimization (HPO) in parallel across many GPUs.
- Improved robustness of DL workflows built with DeepChem and RDKit for ligand-protein pose selection and binding free energy predictions.

Collaboration with Oak Ridge Nat. Lab (ORNL/OLCF) and HZDR Lab (Germany)

(June 2023 – Present)

- PIconGPU (Particle-in-Cell) Center for Accelerated Application Readiness (CAAR): 1 of 8 projects selected in the country to stress test next-gen DOE exascale system
- Ported plasma-physics DL models to the Frontier exascale system at Oak Ridge National Lab (ORNL)
- Improved scalability and optimization of ML models on AMD MI250X GPUs with the ROCm software
- Optimized application by profiling distributed training workloads using AMD profiler Omnitrace
- Ported to NVIDIA's DGX cluster (H100s) for performance and power analysis

PhD Intern- Pacific Northwest National Laboratory's (PNNL) Scalable Analytics and Decision

Optimization group

(June 2022 – August 2022)

- Created a proxy application for scalable Graph Neural Networks (GNNs)
- Created a workflow for sparse, distributed and scalable graphs in distributed memory using Louvain method for community detection

Publications

Journal publications

Gutta V, Ganakammal SR, Jones S, Beyers M, Chandrasekaran S (2024) UNNT: A novel Utility for comparing Neural Net and Tree-based models. PLOS Computational Biology 20(4): e1011504. <https://doi.org/10.1371/journal.pcbi.1011504>

Technical Reports

Oliver Perks, Chongxin, Jayson Falkner, Ricardo Jesus, Prakash Verma, Yueming Hao, Dolan Zhao, Iman Hosseini, John C. Linford, Rivershade, **Vineeth Gutta**, ... Sean Fish. (2021). 2021 A-HUG Hackathon: Cloud Hackathon for Arm-based HPC (1.0). Zenodo. <https://doi.org/10.5281/zenodo.5115938>

Vineeth Gutta (2021). HiT: A framework for increasing portability of deep learning applications in HPC. GitHub. <https://github.com/vgutta/vgutta/blob/main/Prelims.pdf>

Hackathons

2024 UDEL-ATOM Hackathon (**mentor**)

2023 Data Science + AI hackathon University of Delaware (UDEL)

2022 University of Florida GPU Hackathon (**mentor**)

2021 ARM-HPC User Group (A-HUG) Hackathon

Teaching Experience

Graduate teaching assistant

(Aug 2019 – May 2021)

- Intro to Data Mining: Guest lecture twice during the semester in addition to grading assignments for a course with undergraduate and graduate students
- Intro to Software Engineering: helped transition the course to remote format and implemented agile methodologies, such as code reviews, into the curriculum. Taught labs and lead a team of undergraduate and graduate teaching assistants

Instructor for Dept. of Computer and Information Sciences

(June 2020 – Aug 2020)

- Taught, CISC181, a course on object-oriented programming concepts to undergraduate students
- Developed a modified curriculum for fully remote course for summer semester

Invited Talks & Posters

2024 PyTorch Conference (San Francisco): [Lightning talk](#)

- Talk: In-Transit machine learning using PyTorch: A presentation on workflows with science simulations that produce data for online ML model training without storing the data on disk

2024 Supercomputing Conference (Atlanta):

- Talk: **High Performance Python for Science at Scale** workshop
 - o In-Transit Machine Learning of Plasma Simulation on Exascale systems
- Cancer Workshop virtual posters
 - o Machine Learning Surrogate Model for Molecular Pose Optimization in Drug Discovery
 - o UNNT: A novel Utility for comparing Neural Net and Tree-based models

2024 DARWIN Computing Symposium (University of Delaware)

- Talk: DARWIN-driven Innovation: A Showcase of HPC research at Computational Research and Programming Lab

2023 Supercomputing Conference (Denver): Cancer Workshop virtual poster

- Title: Comparison of neural networks with tree-based machine learning approaches for predictive drug response models

Service to Profession

Subreviewer for Supercomputing 2024

Subreviewer for IEEE Cluster 2021

Honors / Awards

Graduate Student Travel Award 2024: \$1200 (University of Delaware)

Mentor: [UD-ATOM Hackathon's](#) Team *Decay Detectives*

- Award category: Technical Complexity
- Computer and Information Sciences Distinguished Graduate Student Award (2021)

Professional Memberships / Affiliations

ACM member since 2015